

Computer Science Exam
C language
2024

Exercise 1: What is the exact output of the following code? (4 points)

```
struct comp {
char address[80];
char phone[11];
} tc;
struct inv {
char barcode[12];
float price;
struct comp *manuf;
};
main()
{
struct inv s;
sprintf(tc.address,"313 Main St.");
strcpy(tc.phone,"3035552479");
s.price=6.42;
strcpy(s.barcode,"1961354128");
s.manuf=&tc;
printf("%c %s\n",s.manuf->address[5],&(tc.phone[6]));
}
```

Answer:

Exercise 2: Write out the memory map for the following code, providing all values at the end of execution. How many total bytes does this code declare for variables? (4 points)

```
struct passport {
char last[10];
int number;
float fees;
double *country_code; };
struct passport list[3],*passenger;
char who[7],a,*b;
double *d,d2;
int i,*j;
d2=888;
i=12;
j=&i;
strcpy(who,"person");
strcpy(list[1].last,"jones");
list[1].number=129783;
list[1].fees=105.37;
list[1].country_code=&d2;
```

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```
passenger=&(list[1]);
b=&(passenger->last[0]);
d=passenger->country_code;
*b='t';
passenger->last[4]='y';
for (i=0; i<5; i++)
    who[i%3]=(char)(*j+i);
```

[illegible]

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Exercise 3: Multiple choices (5 Points)

- 1- Which keyword is used to define a structure in C?
 - A) define
 - B) struct
 - C) typedef
 - D) structdef
- 2- What is the purpose of using malloc() function in C?
 - A) To create a new structure
 - B) To allocate memory dynamically
 - C) To define a new data type
 - D) To initialize an array
- 3- Which of the following search algorithms is based on the principle of divide and conquer?
 - A) Linear search
 - B) Binary search
 - C) Interpolation search
 - D) Depth-first search
- 4- In which sort algorithm is the largest element "bubbled" to the end of the array in each pass?
 - A) Insertion sort
 - B) Selection sort
 - C) Bubble sort
 - D) Merge sort
- 5- How is memory deallocated after using malloc() in C?
 - A) Automatically by the compiler
 - B) Using the free() function
 - C) By setting the pointer to NULL
 - D) By exiting the program
- 6- Which data structure is typically used to implement binary search?
 - A) Array
 - B) Linked list
 - C) Stack
 - D) Queue

7- What will be the output of the following C code?

```
#include <stdio.h>
int main() {
int x = 5;
int *ptr = &x;
*ptr = 10;
printf("%d\n", x);
return 0; }
```

- A) 5
- B) 10
- C) Compilation Error
- D) Undefined Behavior

8- What will be the value of `y` after executing the following code?

```
#include <stdio.h>
int main() {
    int x = 5;
    int *ptr = &x;
    int y = *ptr;
    *ptr = 10;
    printf("%d\n", y);
    return 0; }
```

- A) 5
- B) 10
- C) Garbage Value
- D) Compilation Error

9- What is the output of the following code?

```
#include <stdio.h>
int main() {
    int x = 5;
    int *ptr1 = &x;
    int *ptr2 = ptr1;
    *ptr2 = 10;
    printf("%d\n", x);
    return 0; }
```

- A) 5
- B) 10
- C) Compilation Error
- D) Undefined Behavior

10- What will be the output of the following C code?

```
#include <stdio.h>
int main() {
    int x = 5;
    int *ptr1 = &x;
    int **ptr2 = &ptr1;
    **ptr2 = 10;
    printf("%d\n", x);
    return 0; }
```

- A) 5
- B) 10
- C) Compilation Error
- D) Undefined Behavior

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Exercise 4: (7 points) Consider the following variable declarations. Write code that uses these variables to perform a matrix multiplication. The program should prompt the user for the sizes of the two matrices to be multiplied (from these, it can calculate the size of the result matrix). It should then dynamically allocate memory for the matrices and prompt the user for values for all entries in the matrix. Finally, the program should perform the matrix multiplication and report the result.

```
double **m1,**m2,**mr;
```

```
int m1_rows,m1_cols,m2_rows,m2_cols,mr_rows,mr_cols;
```

Example: if the first matrix is 1×3 with values [1 7 4] and the second matrix is 3×2 with values [1 3], [1 9] and [6 2] in each row, then the result matrix should be 1×2 with values [32 74].